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PDF

Make notes Very detailed (full sentences, no short phrases)
✓ Including all important theoretical points ✓ Including procedural steps wherever steps are given ✓ Formatted cleanly for exam make sure to include each and every imp point

Here are your exam-ready, very detailed notes (full sentences, no short phrases, all important points included, with steps wherever needed) from the uploaded PDF:



AWS Academy Cloud Foundations

Module 8: Databases – Complete Notes

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1. Introduction to Databases in AWS

Module Objectives

After completing this module, a student should be able to:

- Explain Amazon Relational Database Service (Amazon RDS) in detail.
- Identify and describe all functionalities of Amazon RDS.
- Explain Amazon DynamoDB and its working.
- Identify features and use cases of DynamoDB.
- Explain Amazon Redshift and its architecture.
- Explain Amazon Aurora and its benefits.
- Perform practical tasks such as launching, configuring, and interacting with databases in Amazon RDS. AcademyCloudFoundations_Module_...

2. Managed vs Unmanaged Database Services

Question: Differentiate between managed and unmanaged database services.

Unmanaged Services

In unmanaged services, the user is responsible for scaling, fault tolerance, and availability of the system.

The user must manually handle updates, patches, and maintenance.

Requires higher administrative effort and expertise.

Infrastructure setup such as servers, networking, and storage must be handled manually.

Managed Services

In managed services, scaling, fault tolerance, and availability are automatically handled by the service provider.

The service provider automatically manages updates, patches, and maintenance tasks.

Requires minimal administrative effort because most tasks are automated.

Infrastructure setup is abstracted and handled by the cloud provider.

👉 According to the diagram on **page 6**, managed services significantly reduce operational burden by automating key processes.

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3. Challenges of Traditional Relational Databases

Relational databases face several practical challenges:

- Server maintenance increases operational complexity and energy consumption.
- Installation and patching of database software require continuous effort.
- Backup management and ensuring high availability are difficult tasks.
- Scalability is limited due to hardware constraints.
- Data security must be manually implemented and maintained.
- Operating system installation and patching add additional overhead.

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4. Amazon RDS (Relational Database Service)

Definition

Amazon RDS is a managed cloud service that allows users to set up, operate, and scale relational databases easily in the cloud environment.

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4.1 Responsibilities in Amazon RDS

Question: Explain responsibility division in Amazon RDS.

- The user is responsible for application optimization only.
- AWS is responsible for:
 - Operating system installation and patching.
 - Database software installation and patching.
 - Automated database backups.
 - High availability configuration.
 - Scaling of resources.
 - Server maintenance and infrastructure management.

👉 This is clearly illustrated in the **diagram on page 10** showing shared responsibility.

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4.2 Components of Amazon RDS

An Amazon RDS instance consists of:

1. DB Instance Class

- Defines CPU, memory, and network performance.

2. Storage Types

- Magnetic storage.
- General Purpose SSD storage.
- Provisioned IOPS for high performance.

3. Supported Database Engines

- MySQL
- Amazon Aurora
- Microsoft SQL Server
- PostgreSQL
- MariaDB
- Oracle

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4.3 High Availability in RDS (Multi-AZ Deployment)

Working:

- A primary database instance is created in one Availability Zone.
- A standby replica is created in another Availability Zone.
- Data is synchronously replicated between primary and standby.

- In case of failure, automatic failover occurs.

👉 The **diagram on pages 14–15** shows synchronous replication between primary and standby.

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4.4 Read Replicas

Features:

- Uses asynchronous replication.
- Can be promoted to primary instance if required.

Purpose:

- Improves performance for read-heavy applications.
- Offloads read queries from the main database.

👉 The **diagram on page 16** shows primary and replica interaction.

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4.5 Use Cases of Amazon RDS

- Web and mobile applications requiring high throughput and scalability.
- E-commerce platforms requiring secure and cost-effective databases.
- Gaming applications requiring rapid scaling and monitoring.

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4.6 When to Use and When NOT to Use RDS

Use RDS when:

- Complex queries and transactions are required.
- Moderate to high I/O workloads are present.
- High durability is required.

Do NOT use RDS when:

- Extremely high throughput (e.g., 150,000 writes/sec) is needed.
- Sharding is required.
- Simple key-value operations are sufficient (use NoSQL instead).

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4.7 Pricing in Amazon RDS

- Billing is based on clock-hour usage.
- Purchase types include:
 - On-demand instances.
 - Reserved instances (1-year or 3-year commitment).
- Storage charges include:
 - Backup storage (up to 100% free for active DB).
 - Additional storage billed per GB/month.
- Data transfer:
 - Inbound is free.
 - Outbound is charged.

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4.8 Steps: Creating an RDS Database (Lab Procedure)

Question: Explain steps to create and interact with an RDS database.

1. Create a VPC security group to control access.
2. Create a DB subnet group within the VPC.
3. Launch an Amazon RDS DB instance.
4. Configure database parameters such as engine, storage, and instance type.
5. Connect to the database using an application.
6. Perform operations such as querying and updating data.

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5. Amazon DynamoDB (NoSQL Database)

5.1 Relational vs Non-Relational Databases

Relational (SQL)

Data is stored in rows and columns.

Uses fixed schema.

Uses SQL for querying.

Scales vertically.

Non-Relational (NoSQL)

Data is stored as key-value pairs, documents, or graphs.

Uses dynamic schema.

Uses flexible query models.

Scales horizontally.

👉 Example shown on **page 32** compares tabular vs JSON structure.

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5.2 Definition of DynamoDB

Amazon DynamoDB is a fast, flexible NoSQL database service that provides virtually unlimited scalability with low latency performance.

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5.3 Core Components

- Tables: Collection of data.
- Items: Individual records.
- Attributes: Properties of items.
- Primary Keys:
 - Partition key.
 - Composite key (partition + sort key).

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5.4 Key Features of DynamoDB

- Runs entirely on SSD storage.
- Provides single-digit millisecond latency.
- Automatically replicates data across regions.
- Supports key-value and document data models.
- Provides unlimited table size and throughput.
- Highly scalable for modern applications.

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5.5 Partitioning in DynamoDB

- Data is distributed across partitions using partition keys.
- Query operation is efficient when using keys.
- Scan operation is used for searching non-key attributes.

👉 The **diagram on page 35** shows partition-based data distribution.

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6. Amazon Redshift (Data Warehouse)

6.1 Definition

Amazon Redshift is a fully managed data warehouse service designed for analytics and complex queries.

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6.2 Architecture

- Leader node manages query execution.
- Compute nodes perform actual data processing.
- Uses parallel processing for faster results.

👉 The diagram on page 43 shows leader and compute node structure.

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6.3 Features

- Columnar storage improves performance.
- Parallel processing enables high-speed queries.
- Automatic monitoring and scaling.
- Built-in encryption for security.

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6.4 Use Cases

- Enterprise data warehousing.
- Big data analytics.
- SaaS applications requiring analytics.
- Cost-effective large-scale data processing.

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7. Amazon Aurora

7.1 Definition

Amazon Aurora is an enterprise-class relational database that is compatible with MySQL and PostgreSQL.

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7.2 Features

- Fully managed service.
- High performance and scalability.
- Automated backups, patching, and recovery.
- High availability with multiple replicas.
- Multi-layer security.

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7.3 High Availability and Resilience

- Data is replicated across multiple Availability Zones.
- Uses redo logs for recovery.
- Automatically detects and repairs failures.

👉 The **diagrams on pages 52–53** show multi-AZ replication and recovery mechanism.

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8. Choosing the Right Database

Question: How to choose the right AWS database service?

- Use Amazon RDS for relational database requirements.
- Use DynamoDB for NoSQL and high scalability.
- Use Amazon Redshift for analytics and data warehousing.
- Use Amazon Aurora for high-performance relational workloads.
- Use EC2 databases when OS-level control is required.

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9. Case Study Insights

- Large-scale systems may require a combination of relational and NoSQL databases.
- Migration to serverless architectures can involve hybrid solutions.
- High-traffic applications require scalable and high-throughput databases.

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10. Sample Exam Question

Question:

Which of the following is a fully-managed NoSQL database service?

Options:

- A. Amazon RDS
- B. Amazon DynamoDB
- C. Amazon Aurora
- D. Amazon Redshift

Answer:

The correct answer is **Amazon DynamoDB**, because it is a fully managed NoSQL database service.  AcademyCloudFoundations_Module_...

Final Quick Summary (For Revision)

- Amazon RDS is a managed relational database service.
 - DynamoDB is a NoSQL database with unlimited scalability.
 - Redshift is used for data warehousing and analytics.
 - Aurora is a high-performance relational database.
 - Managed services reduce operational burden significantly.
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If you want, I can also:

-  Convert this into **short revision notes (1–2 pages)**
-  Give **important exam questions + answers**
-  Make **comparison charts (RDS vs DynamoDB vs Aurora vs Redshift)**

Sources

